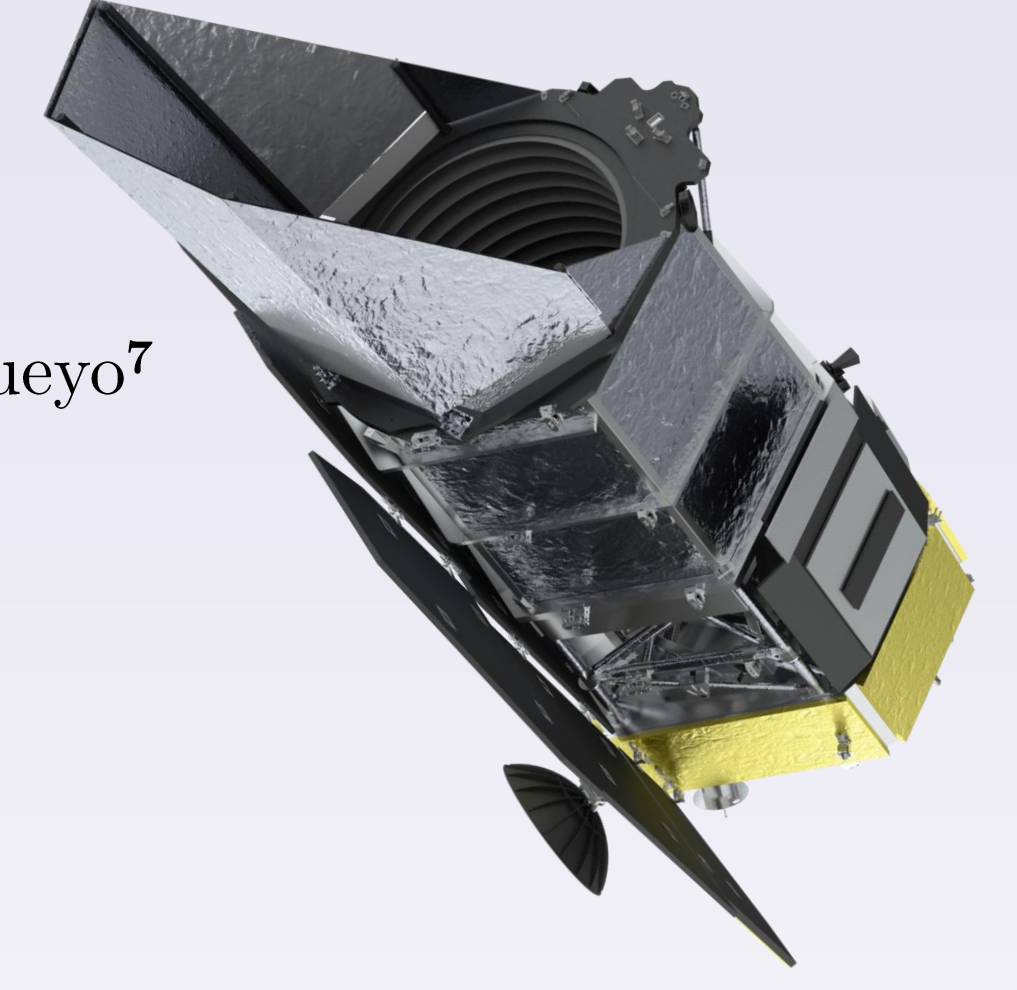


Enhanced wavefront sensing for the Roman Coronagraph Instrument: Gaussian probes and compact model validation



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INTRODUCTION

- Context:** The Roman Space Telescope Coronagraph Instrument prepares the Habitable Worlds Observatory by demonstrating closed-loop focal plane wavefront sensing in space with **Pairwise probing (PWP)**, a technique based on phase diversities introduced by the deformable mirror.

Problem:

Observing faint stars ($V \sim 5$) requires high probe amplitudes. This introduces nonlinear biases that degrade the speckle estimation, depending on the probe shape.

What we did:

Replace nominal probe with Gaussian probe.

This was selected for the first technology demonstrations carried out on Roman in early 2027.

SIMULATION RESULTS

We compare nominal sinc-sinc-sine probes with Gaussian probes. Simulation: Noise-free, conducted at 575 nm, between $3 - 9 \lambda/D$. Focus the estimation only with PWP.

1.

- Find Gaussian width σ_G :** compare STD and PtV of I_{probe} between Gaussian and sinc-sinc-sine.
- STD:** Standard deviation of the intensity for the sinc-sinc-sine probes corresponds to the standard deviation intensity of Gaussian with sigma = 1.2 pitch.
- PtV:** Peak-to-Valley intensity for the sinc-sinc-sine probes corresponds to the Peak-to-Valley intensity of Gaussian with sigma = 0.95 pitch.
- Optimize probe energy in dark hole (DH):** sinc-sinc-sine has been optimized to inject as much light as possible into the DH. Good compromise:
 $\sigma_G = 1$ actuator pitch

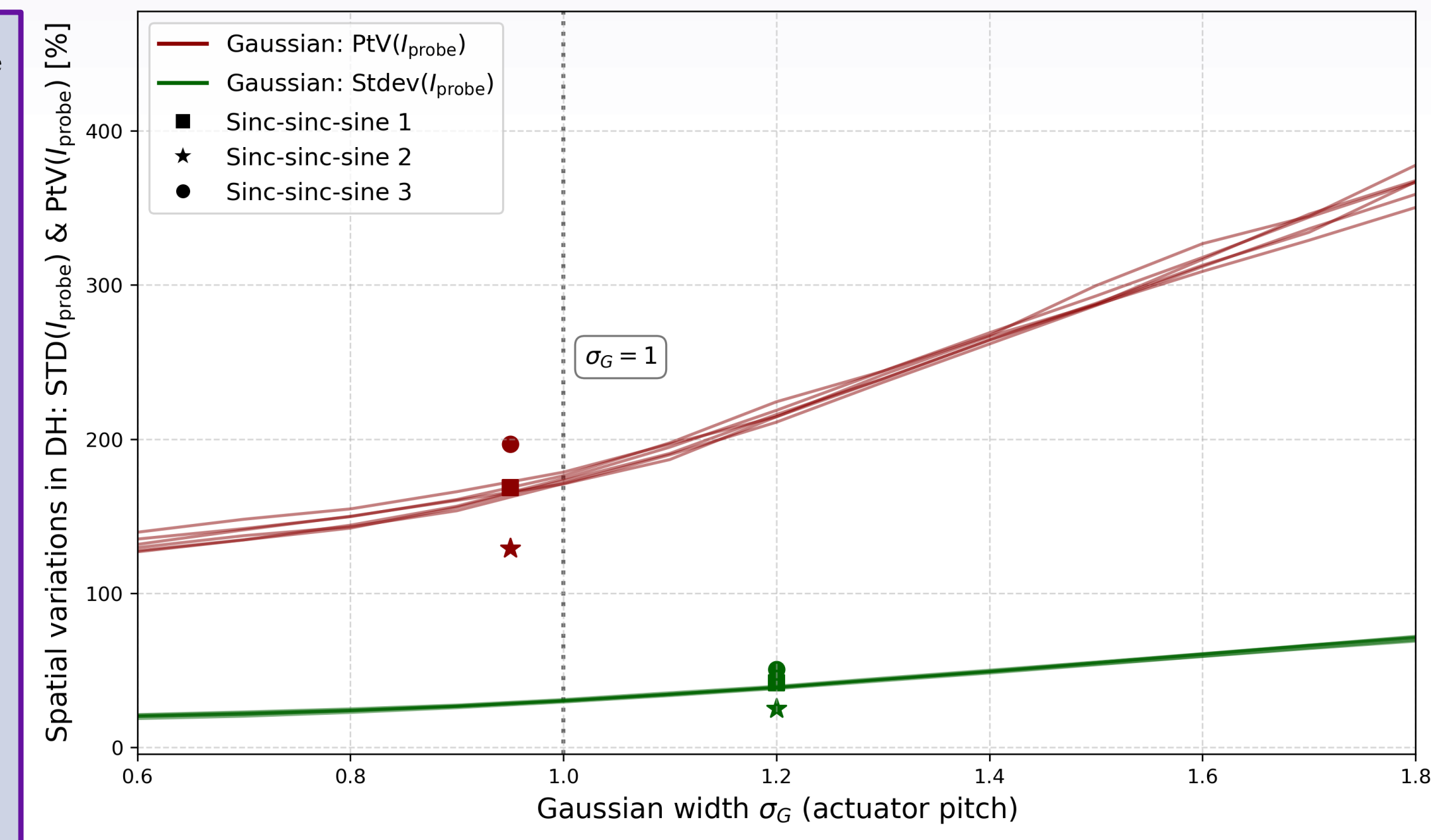


Fig. 3: Asterix simulation of spatial variation (STD & PtV of I_{probe}) in the DH during modulation.

2.

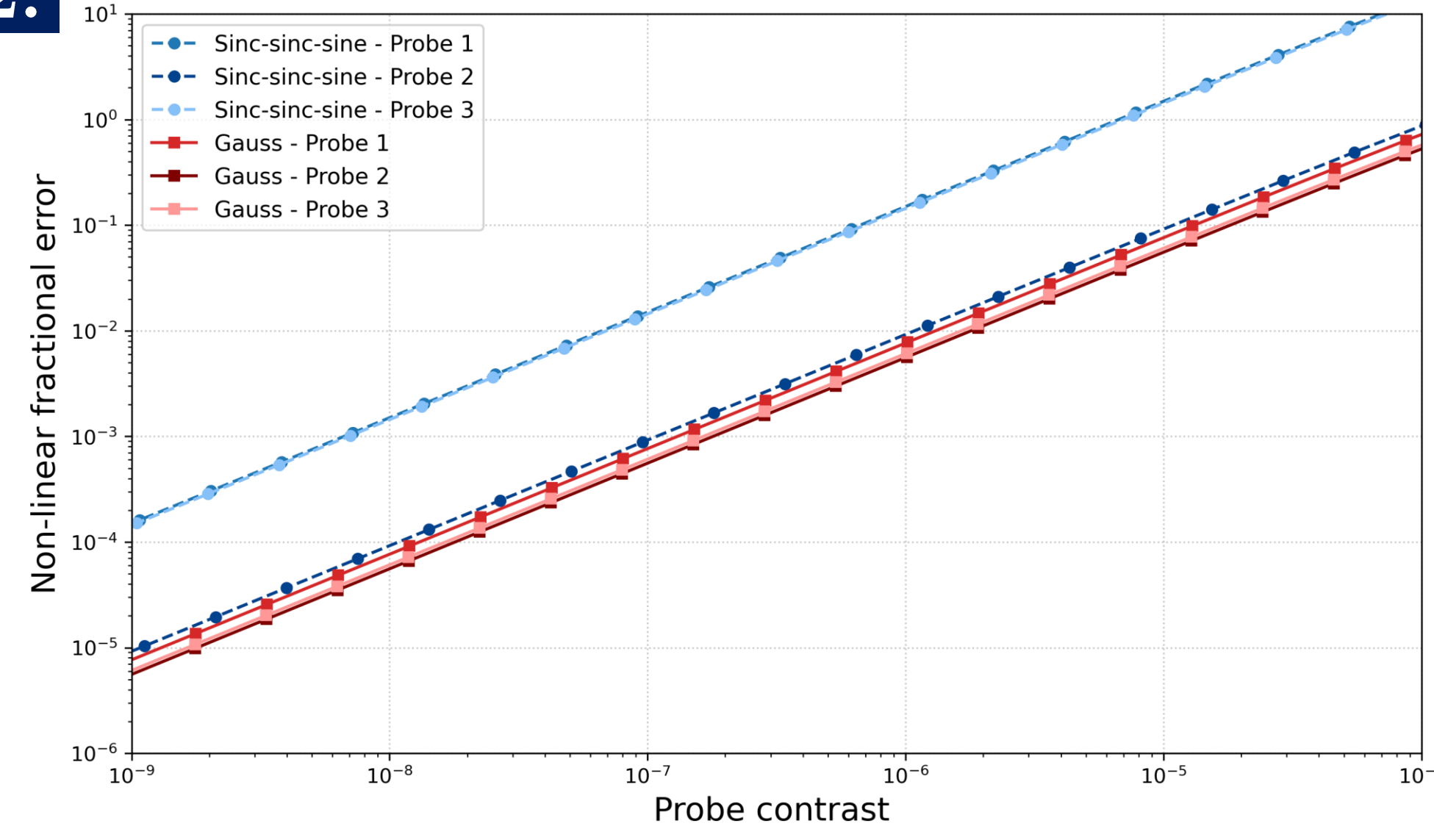


Fig. 4: Asterix simulation of nonlinear error as a function of I_{probe} (DH intensity of 2×10^{-8}).

- Theory:** We simulate nonlinear errors for different probe amplitudes.

- Interpolation:**

$$\text{Non-linear error} \propto I_{\text{probe}}$$

Sinc-sinc-sine probes have much higher errors than the Gaussian probes.

PAIRWISE PROBING

- Goal:** Estimate field of speckles: E_s
- How?** Apply pairs of known probes $\pm \Delta\psi_m$ on the DM.
- Probes intensity measurement $\Delta I_m = I_m^+ - I_m^-$.

- Solve **linear** inverse problem:

$$\underbrace{\Delta I_m}_{\text{Measurement}} = \underbrace{M_{\text{model}}}_{\text{Modulation}} \cdot \underbrace{E_s}_{\text{Speckles}}$$

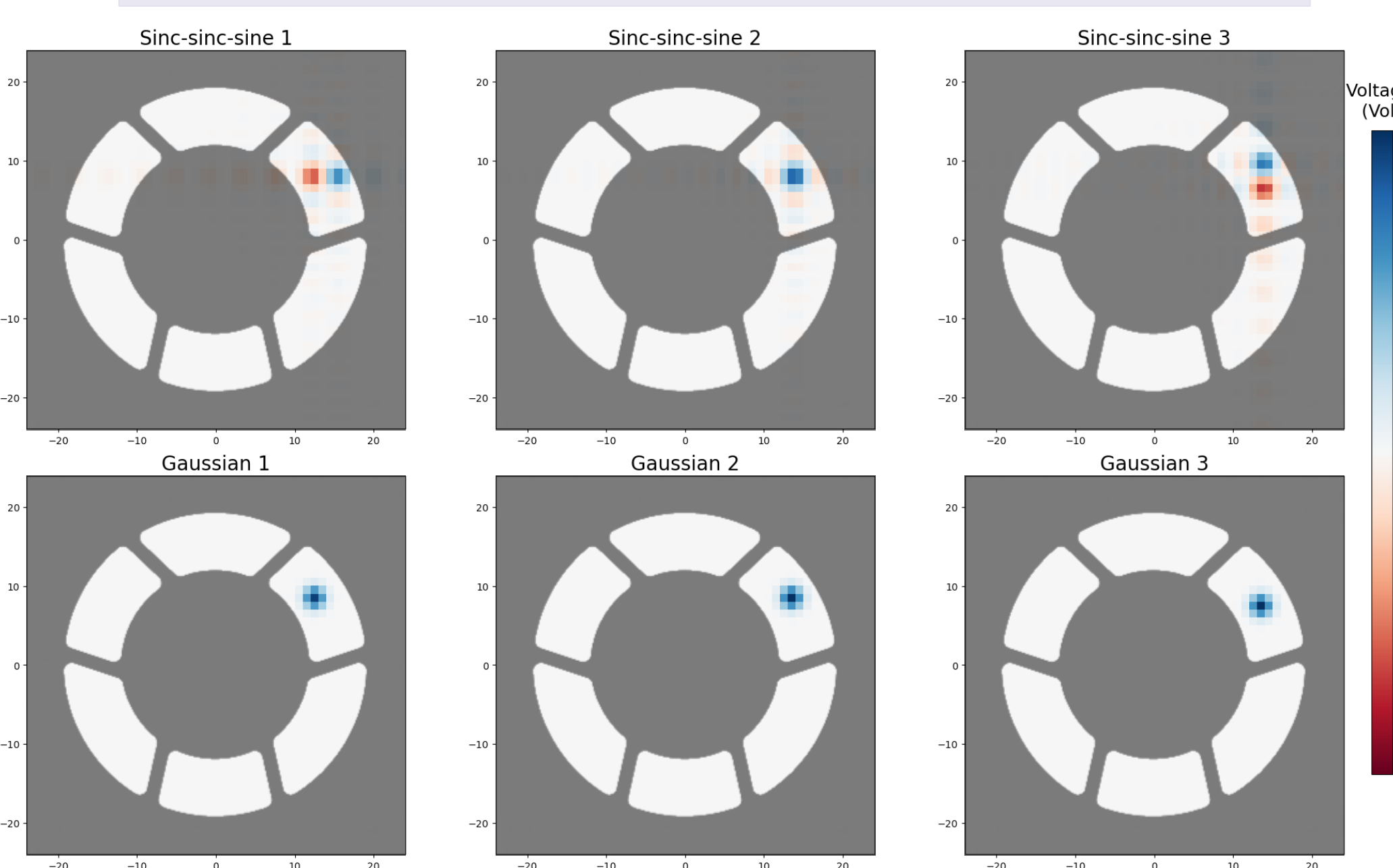


Fig. 2: DM actuator voltages corresponding to the nominal sinc-sinc-sine probes (top) and the alternative Gaussian probes (bottom).

PWP LIMITATIONS

- PWP:** Large probe intensity I_{probe} breaks the linear measurement ΔI_m .
- The dominant nonlinear term depends only on the probe morphology.
- Nonlinear error for a probe $\Delta\psi_m$ = estimation error.
- Errors degrade the PWP algorithm performance.

References: Roman end-to-end simulation: Krist, J. E. *et al.* (2023) | High-order wavefront sensing and control with Roman: Cady, E. *et al.* (2025) | Pairwise probing method: Groff, T. D. *et al.* (2016) | Nonlinear effect with pairwise probing: Laginja, I. *et al.* (2025) | Asterix simulation: Mazoyer, J. *et al.* (2019).

CONCLUSION

- Width of $\sigma_G = 1$ actuator pitch provides good illumination of the DH.
- Using **noise-free simulation** to demonstrate a performance gain of a factor up to 3.5 in the DH with Gaussian probes between $3 - 9 \lambda/D$.
- Gaussian probes** would improve speckles electric field estimate for DH digging for Roman and HWO.

WE SHOULD USE GAUSSIAN PROBES WITH ROMAN AND HWO!

OUTLOOK

- End-to-end performance with corgisim:** Transitioning from the noise-free model to the full model.

PREPARE OUR FIRST ON-SKY TECHNOLOGICAL DEMONSTRATION AT THE BEGINNING OF 2027 WITH ROMAN!



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